

PHOTOREACTIVE COUMARINS FOR PEPTIDE-MEMBRANE INTERACTIONS USING FLUORESCENCE IMAGING AND MASS SPECTROMETRY

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Keywords: Coumarin, photocrosslinking, fluorogenicity, membrane, mass spectrometry

ABSTRACT

Antimicrobial peptides (AMPs) and cell-penetrating peptides (CPPs) are membrane-active molecules with antibacterial and delivery properties. Getting insights into the interactions that underlie their biological activity is crucial to design new sequences with enhanced properties. These weak interactions can be studied by photoaffinity labeling coupled to mass spectrometry (PAL-MS), incorporating the photoreactive moiety on peptidic sequences.^[1-2] This project aims to develop synthetically modified lipids bearing photoreactive moieties on their polar head. Furthermore, we will combine UV photoreactivity and fluorogenicity through the design of original bifunctional hybrids based on coumarins.^[3] Combining spatial localization information with classical PAL-MS methods will allow to better understand peptide behavior.

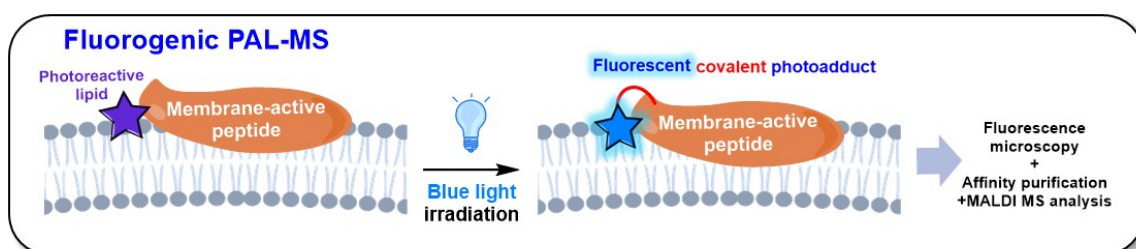


Fig. General representation of the project objectives

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